## Table 1: Descriptive Statistics for New Parameters

### The MEANS Procedure

| Variable | N | Mean | Std Dev | Coeff of Variation | Minimum | Maximum | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|---|
| energy_val | 48 | 17.2525000 | 3.1794690 | 18.4290337 | 13.4200000 | 21.4500000 | 0.1135487 | -1.6296795 |
| depth | 48 | 5.0025000 | 0.3993852 | 7.9837115 | 4.3000000 | 5.4600000 | -0.5617128 | -1.1163351 |
| n_val | 48 | 1.2196563 | 0.1494245 | 12.2513648 | 0.9284421 | 1.4904465 | 0.0474936 | -0.7816312 |

---

## Table 2: Spearman Correlation for Parameter n

### The CORR Procedure

| | |
|---|---|
| 2 With Variables: | energy_val depth |
| 1 Variables: | n_val |

**Simple Statistics**

| Variable | N | Mean | Std Dev | Median | Minimum | Maximum |
|---|---|---|---|---|---|---|
| energy_val | 48 | 17.25250 | 3.17947 | 17.07000 | 13.42000 | 21.45000 |
| depth | 48 | 5.00250 | 0.39939 | 5.07500 | 4.30000 | 5.46000 |
| n_val | 48 | 1.21966 | 0.14942 | 1.21034 | 0.92844 | 1.49045 |

**Spearman Correlation Coefficients, N = 48**
**Prob > |r| under H0: Rho=0**

| | n_val |
|---|---|
| energy_val | -0.74803<br><.0001 |
| depth | 0.63010<br><.0001 |

---

## Table 3: Predicting n from Energy and Depth

### The REG Procedure
### Model: MODEL1
### Dependent Variable: n_val

| | |
|---|---|
| Number of Observations Read | 48 |
| Number of Observations Used | 48 |

**Analysis of Variance**

| Source | DF | Sum of Squares | Mean Square | F Value | Pr > F |
|---|---|---|---|---|---|
| Model | 2 | 1.04117 | 0.52059 | 2846.64 | <.0001 |
| Error | 45 | 0.00823 | 0.00018288 | | |
| Corrected Total | 47 | 1.04940 | | | |

| | | | |
|---|---|---|---|
| Root MSE | 0.01352 | R-Square | 0.9922 |
| Dependent Mean | 1.21966 | Adj R-Sq | 0.9918 |
| Coeff Var | 1.10877 | | |

**Parameter Estimates**

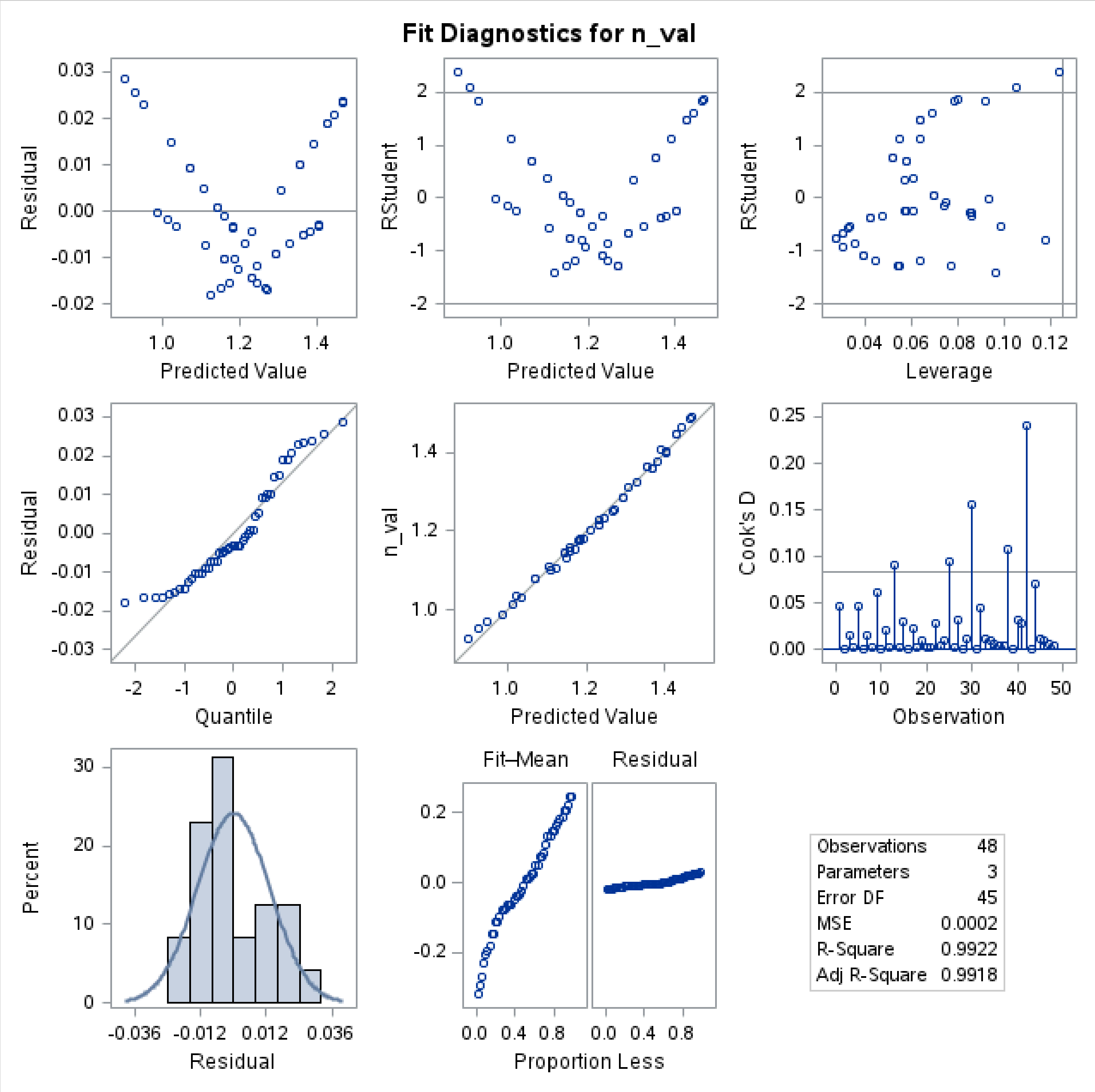
| Variable | DF | Parameter Estimate | Standard Error | t Value | Pr > \|t\| | Variance Inflation |
|---|---|---|---|---|---|---|
| Intercept | 1 | 0.61080 | 0.02700 | 22.63 | <.0001 | 0 |
| energy_val | 1 | -0.03540 | 0.00062041 | -57.07 | <.0001 | 1.00000 |
| depth | 1 | 0.24381 | 0.00494 | 49.36 | <.0001 | 1.00000 |

| Collinearity Diagnostics | | | | | |
|---|---|---|---|---|---|
| | | | Proportion of Variation | | |
| Number | Eigenvalue | Condition Index | Intercept | energy_val | depth |
| 1 | 2.97428 | 1.00000 | 0.00058839 | 0.00358 | 0.00069671 |
| 2 | 0.02278 | 11.42739 | 0.02685 | 0.93755 | 0.05979 |
| 3 | 0.00295 | 31.76991 | 0.97256 | 0.05887 | 0.93952 |

---

## Table 3: Predicting n from Energy and Depth

**The REG Procedure**
**Model: MODEL1**
**Dependent Variable: n_val**




Residual by Regressors for n_val
Residual
0.03
0.02
0.01
0.00
-0.01
-0.02
14
16
18
20
22
energy_val
4.25
4.50
4.75
5.00
5.25
5.50
depth

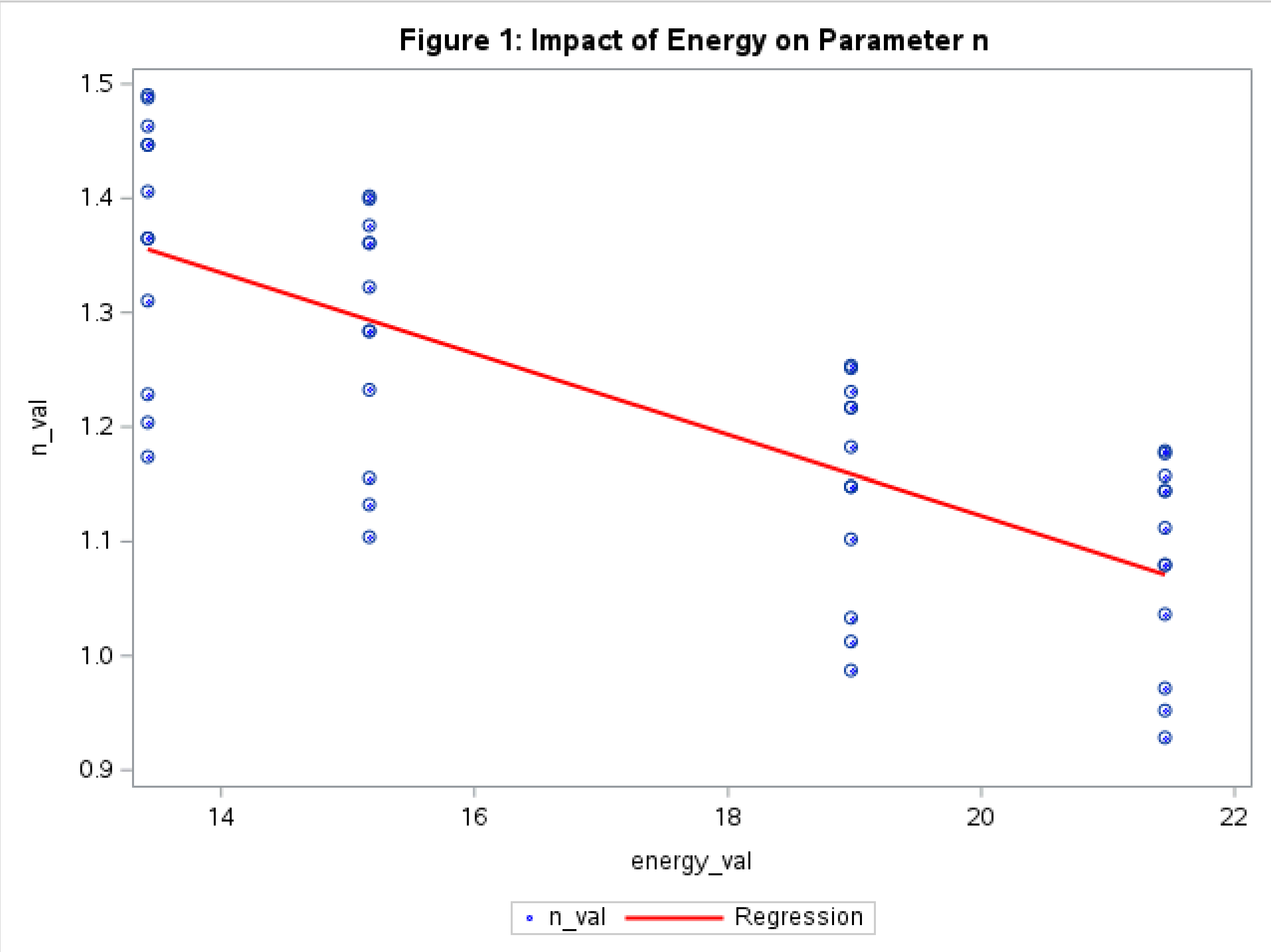
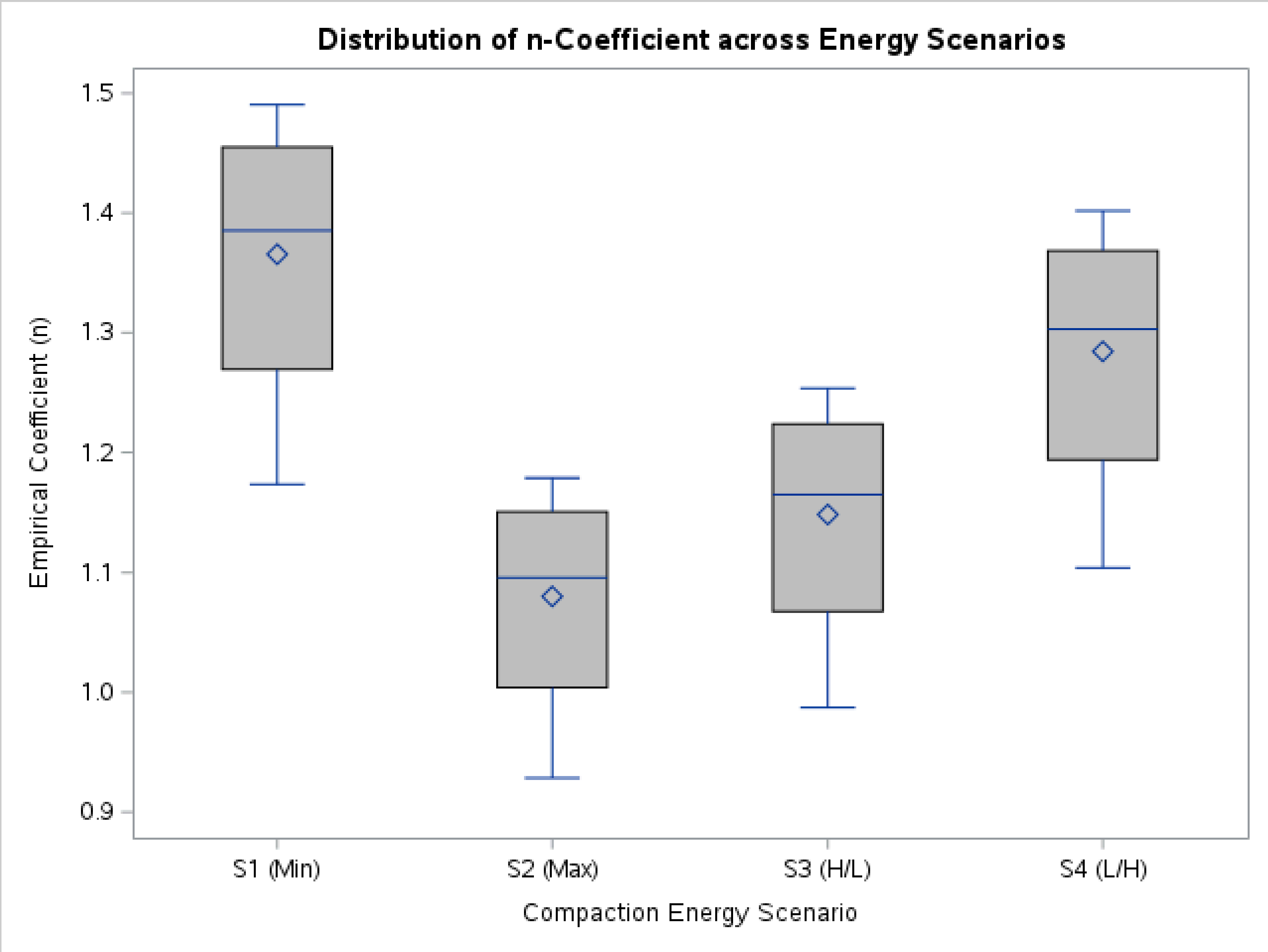
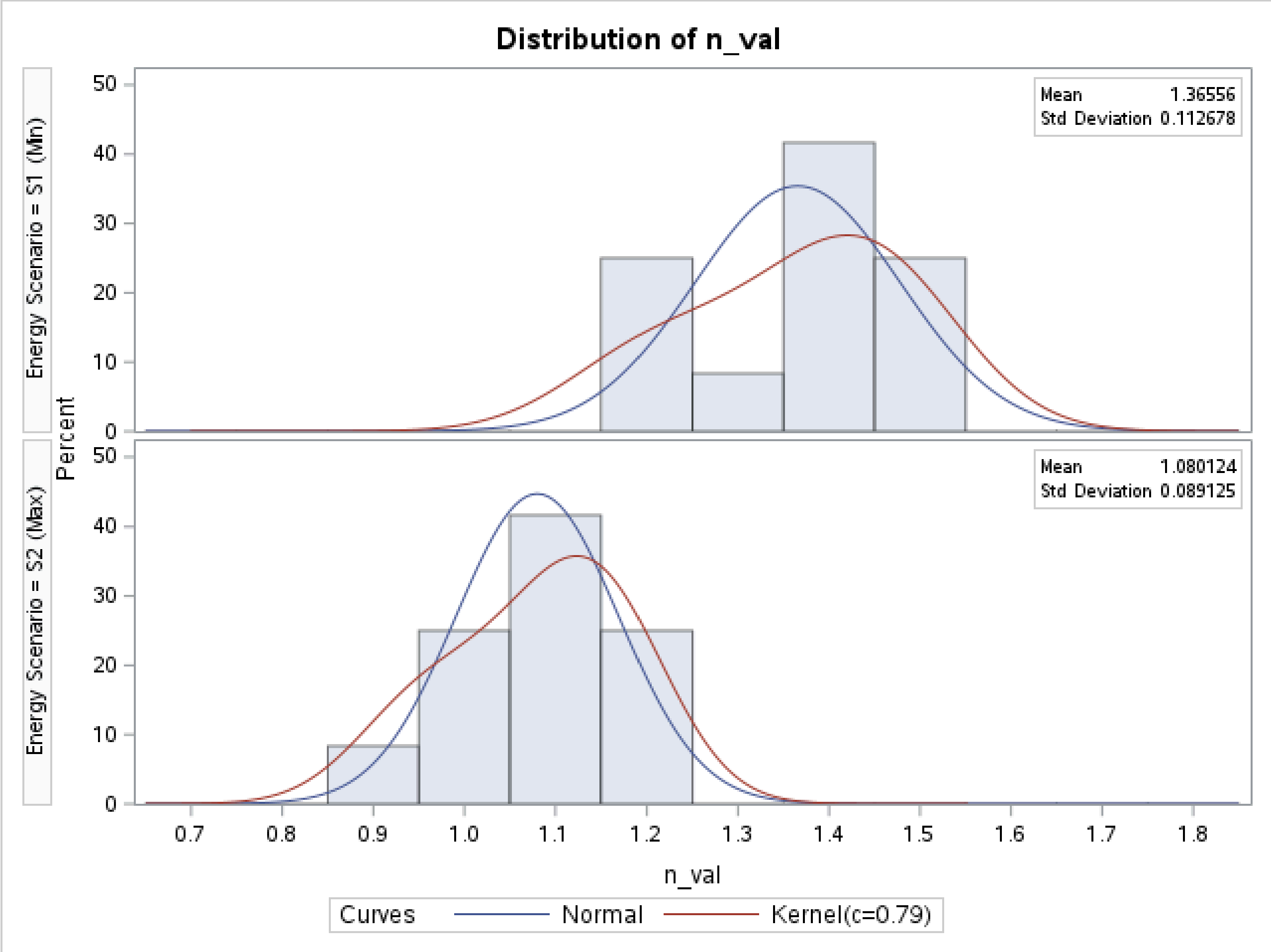


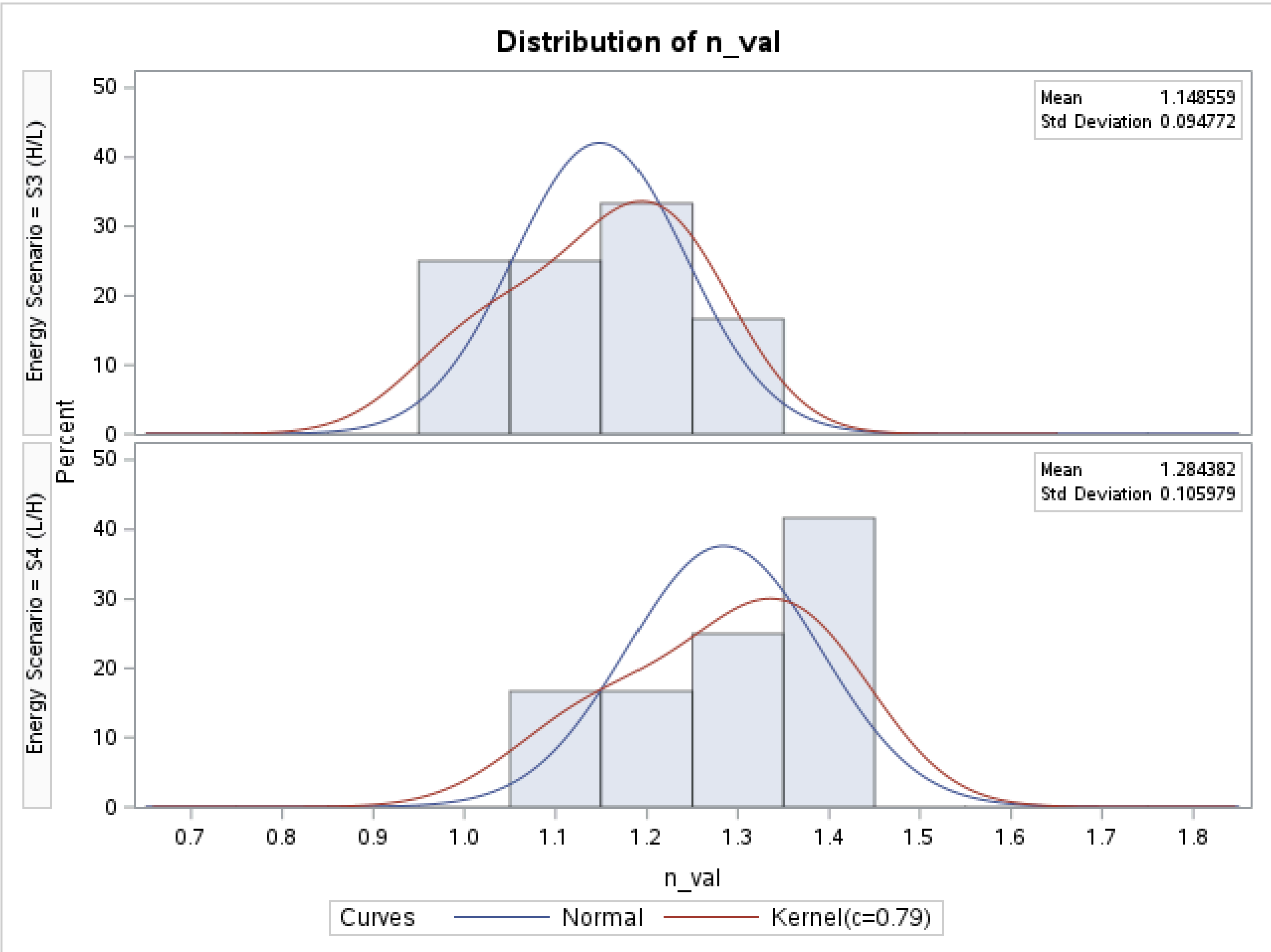
Figure 1: Impact of Energy on Parameter n
n_val
1.5
1.4
1.3
1.2
1.1
1.0
0.9
14
16
18
20
22
energy_val
n_val
Regression




## Probability Density of n for each Energy Input

### The UNIVARIATE Procedure




**Probability Density of n for each Energy Input**

**The UNIVARIATE Procedure**

Energy Scenario = S1 (Min)
Fitted Normal Distribution for n_val

| Parameters for Normal Distribution | | |
|---|---|---|
| **Parameter** | **Symbol** | **Estimate** |
| **Mean** | Mu | 1.36556 |
| **Std Dev** | Sigma | 0.112678 |

| Goodness-of-Fit Tests for Normal Distribution | | | | |
|---|---|---|---|---|
| **Test** | **Statistic** | | **p Value** | |
| **Kolmogorov-Smirnov** | **D** | 0.18112876 | **Pr > D** | >0.150 |
| **Cramer-von Mises** | **W-Sq** | 0.07681524 | **Pr > W-Sq** | 0.215 |
| **Anderson-Darling** | **A-Sq** | 0.49604152 | **Pr > A-Sq** | 0.179 |

| Quantiles for Normal Distribution | | |
|---|---|---|
| | **Quantile** | |
| **Percent** | **Observed** | **Estimated** |
| **1.0** | 1.17379 | 1.10343 |
| **5.0** | 1.17379 | 1.18022 |
| **10.0** | 1.20382 | 1.22116 |
| **25.0** | 1.26934 | 1.28956 |
| **50.0** | 1.38535 | 1.36556 |
| **75.0** | 1.45496 | 1.44156 |
| **90.0** | 1.48772 | 1.50996 |
| **95.0** | 1.49045 | 1.55090 |
| **99.0** | 1.49045 | 1.62769 |

---

## Probability Density of n for each Energy Input

The UNIVARIATE Procedure
Energy Scenario = S2 (Max)
Fitted Normal Distribution for n_val

| Parameters for Normal Distribution | | |
|---|---|---|
| **Parameter** | **Symbol** | **Estimate** |
| **Mean** | Mu | 1.080124 |
| **Std Dev** | Sigma | 0.089125 |

| Goodness-of-Fit Tests for Normal Distribution | | | | |
|---|---|---|---|---|
| **Test** | **Statistic** | | **p Value** | |
| **Kolmogorov-Smirnov** | **D** | 0.18112876 | **Pr > D** | >0.150 |
| **Cramer-von Mises** | **W-Sq** | 0.07681524 | **Pr > W-Sq** | 0.215 |
| **Anderson-Darling** | **A-Sq** | 0.49604152 | **Pr > A-Sq** | 0.179 |

| Quantiles for Normal Distribution | | |
|---|---|---|
| | **Quantile** | |
| **Percent** | **Observed** | **Estimated** |
| **1.0** | 0.92844 | 0.87279 |
| **5.0** | 0.92844 | 0.93353 |
| **10.0** | 0.95219 | 0.96590 |
| **25.0** | 1.00401 | 1.02001 |
| **50.0** | 1.09578 | 1.08012 |
| **75.0** | 1.15084 | 1.14024 |
| **90.0** | 1.17675 | 1.19434 |
| **95.0** | 1.17891 | 1.22672 |

| Quantiles for Normal Distribution | | |
|---|---|---|
| | Quantile | |
| Percent | Observed | Estimated |
| 99.0 | 1.17891 | 1.28746 |

## Probability Density of n for each Energy Input

The UNIVARIATE Procedure
Energy Scenario = S3 (H/L)
Fitted Normal Distribution for n_val

| Parameters for Normal Distribution | | |
|---|---|---|
| Parameter | Symbol | Estimate |
| Mean | Mu | 1.148559 |
| Std Dev | Sigma | 0.094772 |

| Goodness-of-Fit Tests for Normal Distribution | | | | |
|---|---|---|---|---|
| Test | Statistic | | p Value | |
| Kolmogorov-Smirnov | D | 0.18112876 | Pr > D | >0.150 |
| Cramer-von Mises | W-Sq | 0.07681524 | Pr > W-Sq | 0.215 |
| Anderson-Darling | A-Sq | 0.49604152 | Pr > A-Sq | 0.179 |

| Quantiles for Normal Distribution | | |
|---|---|---|
| | Quantile | |
| Percent | Observed | Estimated |
| 1.0 | 0.98727 | 0.92809 |
| 5.0 | 0.98727 | 0.99267 |
| 10.0 | 1.01252 | 1.02710 |
| 25.0 | 1.06763 | 1.08464 |
| 50.0 | 1.16521 | 1.14856 |
| 75.0 | 1.22375 | 1.21248 |
| 90.0 | 1.25130 | 1.27001 |
| 95.0 | 1.25360 | 1.30445 |
| 99.0 | 1.25360 | 1.36903 |

## Probability Density of n for each Energy Input

The UNIVARIATE Procedure
Energy Scenario = S4 (L/H)
Fitted Normal Distribution for n_val

| Parameters for Normal Distribution | | |
|---|---|---|
| Parameter | Symbol | Estimate |
| Mean | Mu | 1.284382 |
| Std Dev | Sigma | 0.105979 |

| Goodness-of-Fit Tests for Normal Distribution | | | | |
|---|---|---|---|---|
| Test | Statistic | | p Value | |
| Kolmogorov-Smirnov | D | 0.18112876 | Pr > D | >0.150 |
| Cramer-von Mises | W-Sq | 0.07681524 | Pr > W-Sq | 0.215 |
| Anderson-Darling | A-Sq | 0.49604152 | Pr > A-Sq | 0.179 |

| Quantiles for Normal Distribution | | |
|---|---|---|
| | Quantile | |
| Percent | Observed | Estimated |
| **1.0** | 1.10402 | 1.03784 |
| **5.0** | 1.10402 | 1.11006 |
| **10.0** | 1.13226 | 1.14856 |
| **25.0** | 1.19388 | 1.21290 |
| **50.0** | 1.30300 | 1.28438 |
| **75.0** | 1.36847 | 1.35586 |
| **90.0** | 1.39928 | 1.42020 |
| **95.0** | 1.40184 | 1.45870 |
| **99.0** | 1.40184 | 1.53093 |